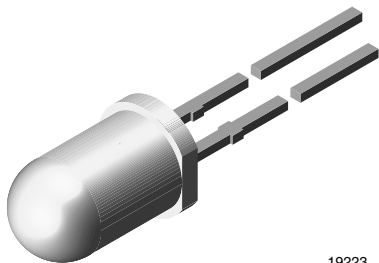


Ultrabright White LED, Ø 5 mm Untinted Non-Diffused Package



19223

DESCRIPTION

The VLHW5100 is a clear, non-diffused 5 mm LED for high end applications where supreme luminous intensity required.

These lamps with clear untinted plastic case utilize the highly developed ultrabright InGaN technologies.

The lens and the viewing angle is optimized to achieve best performance of light output and visibility.

PRODUCT GROUP AND PACKAGE DATA

- Product group: LED
- Package: 5 mm
- Product series: standard
- Angle of half intensity: $\pm 10^\circ$

FEATURES

- Untinted non-diffused lens
- Utilizing ultrabright InGaN technology
- High luminous intensity
- Luminous intensity and color categorized for each packing unit
- ESD-withstand voltage: up to 4 kV according to JESD22-A114-B
- Circuit protection by Zener diode
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912



RoHS
COMPLIANT
HALOGEN
FREE
GREEN
(5-2008)

APPLICATIONS

- Interior and exterior lighting
- Outdoor LED panels
- Instrumentation and front panel indicators
- Replaces incandescent lamps
- Light guide compatible

PARTS TABLE

PART	COLOR	LUMINOUS INTENSITY (mcd)			at I _F (mA)	COORDINATE (x, y)			at I _F (mA)	FORWARD VOLTAGE (V)			at I _F (mA)	TECHNOLOGY
		MIN.	TYP.	MAX.		MIN.	TYP.	MAX.		MIN.	TYP.	MAX.		
VLHW5100	White	5600	-	11 200	20	-	0.33, 0.33	-	20	2.8	-	3.6	20	InGaN and converter

ABSOLUTE MAXIMUM RATINGS (T_{amb} = 25 °C, unless otherwise specified)

PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
Reverse voltage		V _R	5	V
DC forward current		I _F	30	mA
Peak forward current	at 1 kHz, t _p /T = 0.1	I _{FSM}	0.1	A
Power dissipation		P _V	100	mW
Zener reverse current		I _Z	100	mA
Junction temperature		T _j	100	°C
Operating temperature range		T _{amb}	-40 to +100	°C
Storage temperature range		T _{stg}	-40 to +100	°C
Soldering temperature	t ≤ 5 s	T _{sd}	260	°C
Thermal resistance junction-to-ambient		R _{thJA}	400	K/W

**OPTICAL AND ELECTRICAL CHARACTERISTICS** ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)
WHITE VLHW5100

PARAMETER	TEST CONDITION	PART	SYMBOL	MIN.	TYP.	MAX.	UNIT
Luminous intensity	$I_F = 20\text{ mA}$	VLHW5100	I_V	5600	-	11 200	mcd
Chromaticity coordinate x acc. to CIE 1931	$I_F = 20\text{ mA}$		x	-	0.33	-	
Chromaticity coordinate y acc. to CIE 1931	$I_F = 20\text{ mA}$		y	-	0.33	-	
Angle of half intensity	$I_F = 20\text{ mA}$		ϕ	-	± 10	-	$^{\circ}$
Forward voltage	$I_F = 20\text{ mA}$		V_F	2.8	-	3.6	V
Reverse current	$V_R = 5\text{ V}$		I_R	-	-	50	μA
Temperature coefficient of V_F	$I_F = 20\text{ mA}$		TC_{VF}	-	-4	-	mV/K
Temperature coefficient of I_V	$I_F = 20\text{ mA}$		TC_{IV}	-	-0.5	-	% / K

CHROMATICITY COORDINATED CLASSIFICATION

GROUP	X		Y	
	MIN.	MAX.	MIN.	MAX.
3A	0.2900	0.3025	$y = 1.4x - 0.121$	$y = 1.4x - 0.071$
3B	0.3025	0.3150	$y = 1.4x - 0.121$	$y = 1.4x - 0.071$
3C	0.2900	0.3025	$y = 1.4x - 0.171$	$y = 1.4x - 0.121$
3D	0.3025	0.3150	$y = 1.4x - 0.171$	$y = 1.4x - 0.121$
4A	0.3150	0.3275	$y = 1.4x - 0.121$	$y = 1.4x - 0.071$
4B	0.3275	0.3400	$y = 1.4x - 0.121$	$y = 1.4x - 0.071$
4C	0.3150	0.3275	$y = 1.4x - 0.171$	$y = 1.4x - 0.121$
4D	0.3275	0.3400	$y = 1.4x - 0.171$	$y = 1.4x - 0.121$
5A	0.3400	0.3525	$y = 1.4x - 0.121$	$y = 1.4x - 0.071$
5B	0.3525	0.3650	$y = 1.4x - 0.121$	$y = 1.4x - 0.071$
5C	0.3400	0.3525	$y = 1.4x - 0.171$	$y = 1.4x - 0.121$
5D	0.3525	0.3650	$y = 1.4x - 0.171$	$y = 1.4x - 0.121$

Note

- Chromaticity coordinate groups are tested with a tolerance of ± 0.01

LUMINOUS INTENSITY CLASSIFICATION

GROUP	LIGHT INTENSITY (mcd)	
	MIN.	MAX.
DB	5600	7100
EA	7100	9000
EB	9000	11 200

Note

- Luminous intensity is tested with an accuracy of $\pm 11\%$.
The above type numbers represent the order groups which include only a few brightness groups. Only one group will be shipped on each reel (there will be no mixing of two groups on each reel). In order to ensure availability, single brightness groups will not be orderable.
In a similar manner for colors where color groups are measured and binned, single color groups will be shipped on any one reel. In order to ensure availability, single color groups will not be orderable

FORWARD VOLTAGE CLASSIFICATION

GROUP	FORWARD VOLTAGE (V)	
	MIN.	MAX.
0	2.8	3.0
1	3.0	3.2
2	3.2	3.4
3	3.4	3.6

Note

- Forward voltage is tested with an accuracy of $\pm 0.1\text{ V}$

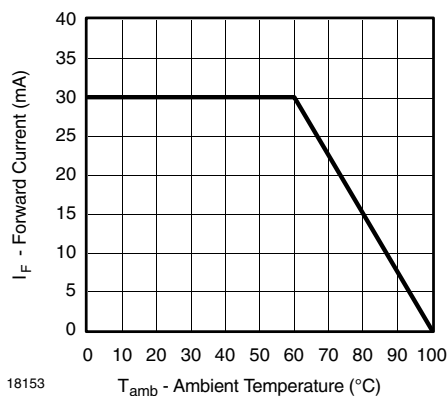
TYPICAL CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)


Fig. 1 - Forward Current vs. Ambient Temperature

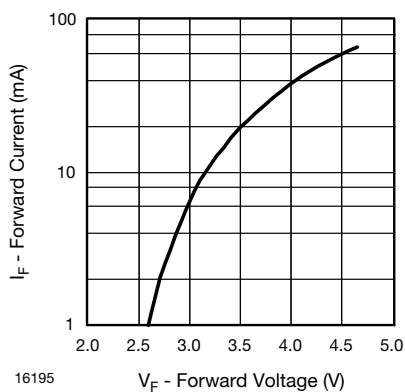


Fig. 4 - Forward Current vs. Forward Voltage

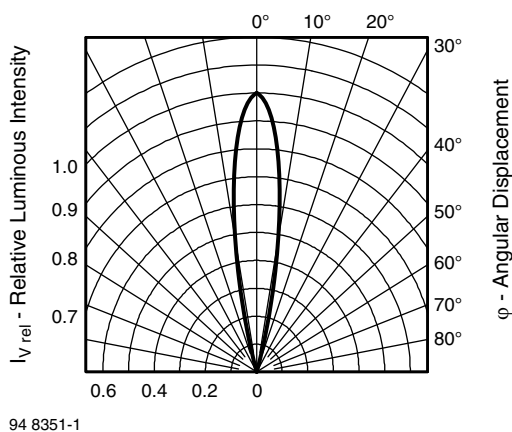


Fig. 2 - Relative Luminous Intensity vs. Angular Displacement

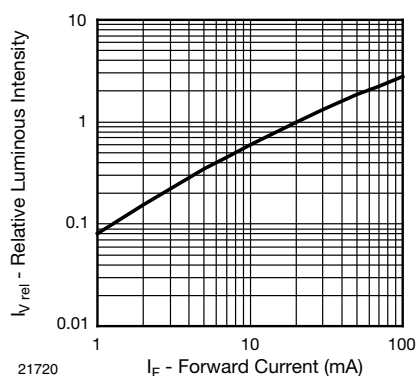


Fig. 5 - Relative Luminous Flux vs. Forward Current

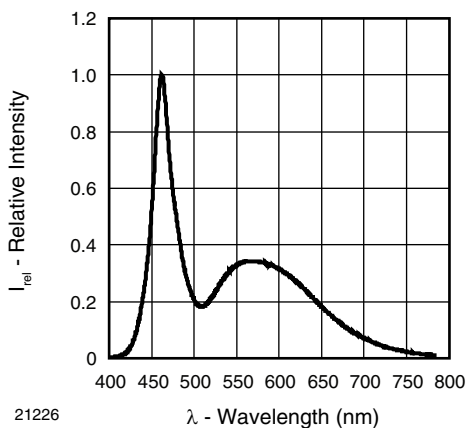


Fig. 3 - Relative Intensity vs. Wavelength

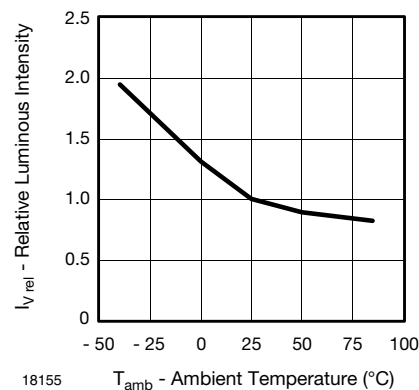


Fig. 6 - Relative Luminous Intensity vs. Ambient Temperature

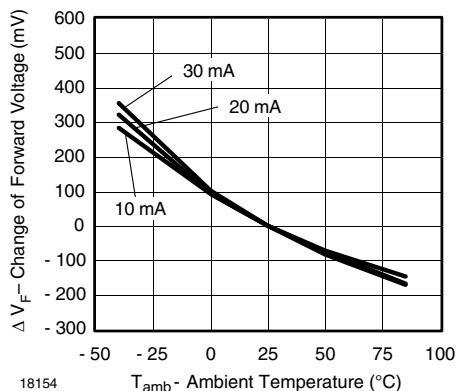


Fig. 7 - Change of Forward Voltage vs. Ambient Temperature

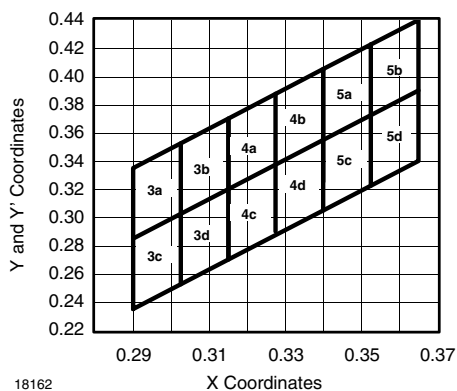


Fig. 8 - Coordinates of Colorgroups

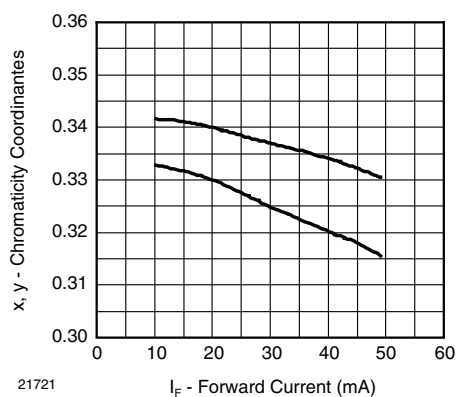
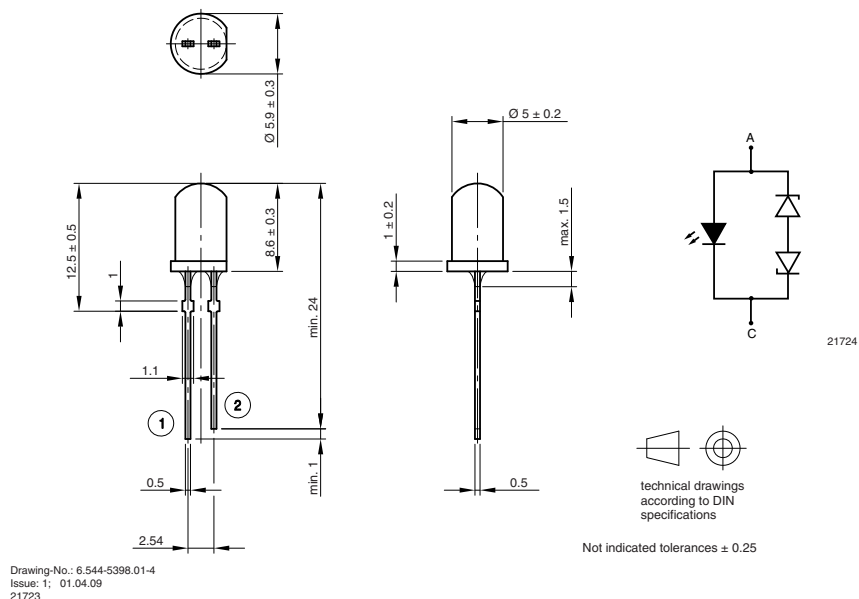
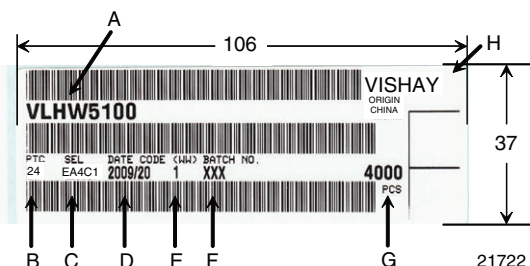


Fig. 9 - Chromaticity Coordinate Shift vs. Forward Current

PACKAGE DIMENSIONS in millimeters

BAR CODE PRODUCT LABEL


- A) Type of component
- B) Manufacturing plant
- C) SEL - selection code (bin):
e.g.: EA = code for luminous intensity group
4C = code for chromaticity coordinate
1 = code for forward voltage
- D) Date code year / week
- E) Day code (e.g. 1: Monday)
- F) Batch no.
- G) Total quantity
- H) Company code



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